Design a fully connected neural network with 3 input neurons, 3 hidden layers, and 1 output neuron. You may choose the number of neurons in each hidden layer and specify the activation functions used.

1. Define the architecture clearly, including the dimensions of all weight matrices and bias vectors.
2. Initialize all weights and biases with small numerical values.
3. Perform the complete forward pass for one training example and calculate the output step by step.
4. Define a suitable loss function (for example, Mean Squared Error).
5. Derive all gradients manually using backpropagation.
6. Compute the gradients for each weight and bias in every layer.
7. Show the complete backward pass, explaining how the error flows from the output layer back to the first hidden layer.
8. Perform one full parameter update using gradient descent and show the updated weights.